

# Docker Desktop Reset Workflow and Container Management

## Docker Desktop Reset Workflow and Container Management

This document explains what happens when you stop or delete containers in Docker Desktop, and how to safely reset your stack using Docker Compose.

---

### 1. Stopping a Container

- Stopping car-e-pool-api halts only that container.
- The MySQL container (car-e-pool-mysql) continues running unless stopped separately.
- Your API (Spring Boot app) becomes inaccessible on port 8080.
- Networks and volumes remain intact, so restarting reconnects to the same database without data loss.



Equivalent to docker-compose stop for that container.

---

### 2. Stop vs Down

#### docker-compose stop

- Stops containers but keeps definitions, networks, and volumes.
- Containers can be restarted quickly with docker-compose start.

#### docker-compose down

- Stops and removes containers and networks.
- Volumes remain unless you add -v.

#### docker-compose down -v

- Stops and removes containers, networks, and volumes.
  - Deletes database data stored in volumes.
-

### 3. Docker Desktop Buttons

- **Stop button** → Equivalent to docker-compose stop.
  - **Delete button** → Equivalent to docker-compose down.
  - **Delete + Remove volumes** → Equivalent to docker-compose down -v.
- 

### 4. Safe Reset Workflow

#### Step 1: Stop containers (optional)

```
docker-compose stop
```

Halts containers without removing anything.

#### Step 2: Tear down and wipe volumes

```
docker-compose down -v
```

Removes containers, networks, and volumes. Database data is erased.

#### Step 3: Rebuild and start fresh

```
docker-compose up -d --build
```

Recreates containers, networks, and volumes. MySQL initializes with a fresh database.

#### Step 4: Verify

```
docker-compose ps
```

Shows running containers (API + MySQL).

---

### 5. Best Practices

- **Daily workflow:** Use stop when pausing work to preserve data.
  - **Reset workflow:** Use down when you want a fresh environment but keep DB data.
  - **Full reset:** Use down -v when you want to wipe all data and start from scratch.
  - **Production caution:** Avoid -v unless you intentionally want to drop all data.
- 

This guide helps you understand the impact of stopping, deleting, and resetting containers in Docker Desktop, ensuring you can manage your local development environment safely.